

18/04/2025



Aakash

Medical | IIT-JEE | Foundations

Corporate Office: AESL, 3rd Floor, Incuspaze Campus-2, Plot-13,
Sector-18, Udyog Vihar, Gurugram, Haryana-122015

CODE-A



AIM - 720

(Advanced **INTENSIVE** Mastery for 720)

MM : 720

CST-7

Time : 180 Mins.

Answers

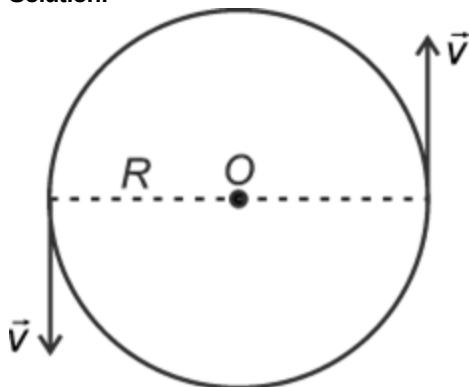
1. (3)	37. (3)	73. (3)	109. (2)	145. (2)
2. (1)	38. (3)	74. (1)	110. (3)	146. (4)
3. (1)	39. (3)	75. (1)	111. (4)	147. (4)
4. (2)	40. (4)	76. (1)	112. (4)	148. (4)
5. (2)	41. (1)	77. (4)	113. (3)	149. (2)
6. (2)	42. (3)	78. (4)	114. (2)	150. (2)
7. (4)	43. (2)	79. (3)	115. (1)	151. (1)
8. (3)	44. (4)	80. (1)	116. (1)	152. (2)
9. (3)	45. (1)	81. (2)	117. (3)	153. (3)
10. (4)	46. (4)	82. (1)	118. (3)	154. (1)
11. (3)	47. (3)	83. (1)	119. (2)	155. (1)
12. (3)	48. (2)	84. (4)	120. (3)	156. (4)
13. (3)	49. (4)	85. (3)	121. (4)	157. (2)
14. (2)	50. (2)	86. (1)	122. (1)	158. (2)
15. (2)	51. (3)	87. (1)	123. (3)	159. (4)
16. (4)	52. (2)	88. (2)	124. (2)	160. (1)
17. (2)	53. (3)	89. (3)	125. (2)	161. (3)
18. (4)	54. (1)	90. (4)	126. (4)	162. (3)
19. (3)	55. (2)	91. (2)	127. (3)	163. (2)
20. (2)	56. (2)	92. (3)	128. (4)	164. (1)
21. (3)	57. (3)	93. (2)	129. (4)	165. (2)
22. (3)	58. (2)	94. (4)	130. (2)	166. (3)
23. (1)	59. (4)	95. (4)	131. (1)	167. (3)
24. (2)	60. (1)	96. (2)	132. (4)	168. (3)
25. (2)	61. (4)	97. (2)	133. (2)	169. (1)
26. (2)	62. (3)	98. (3)	134. (2)	170. (3)
27. (3)	63. (2)	99. (2)	135. (3)	171. (3)
28. (3)	64. (4)	100. (3)	136. (3)	172. (4)
29. (2)	65. (3)	101. (3)	137. (2)	173. (4)
30. (2)	66. (4)	102. (2)	138. (4)	174. (3)
31. (2)	67. (1)	103. (3)	139. (4)	175. (3)
32. (2)	68. (2)	104. (3)	140. (4)	176. (3)
33. (1)	69. (4)	105. (1)	141. (2)	177. (1)
34. (2)	70. (3)	106. (1)	142. (2)	178. (3)
35. (3)	71. (4)	107. (3)	143. (4)	179. (1)
36. (4)	72. (2)	108. (2)	144. (3)	180. (1)

Hints and Solutions

PHYSICS

(1) Answer : (3)

Solution:

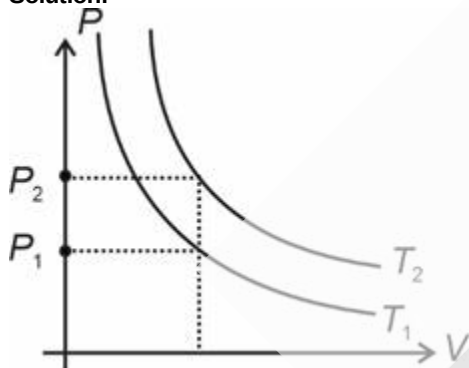


$$|\Delta \vec{v}| = |v\hat{j} - (-v\hat{j})|$$

$$= 2v$$

(2) Answer : (1)

Solution:



At constant volume, $P_2 > P_1$
 $\Rightarrow T_2 > T_1$

(3) Answer : (1)

Solution:

For an adiabatic expansion, $\Delta Q = 0$ and $\Delta U \neq 0$.
 Heat and work are considered to be path functions.
 $\eta_{\text{engine}} \leq \eta_{\text{carnot}}$

(4) Answer : (2)

Solution:

$$\Delta P = \frac{4S}{r} \Rightarrow \Delta P \propto \frac{1}{r}$$

$$\Rightarrow 2\Delta P \times r_A = \Delta P \times r_B$$

$$\Rightarrow \frac{r_A}{r_B} = \frac{1}{2} \Rightarrow \frac{V_A}{V_B} = \frac{1}{8}$$

(5) Answer : (2)

Solution:

An object can also float due to surface tension. The contact angle between water and glass is acute.

(6) Answer : (2)

Solution:

$$R_{\text{eq}} = R_1 + R_2$$

$$\Rightarrow \frac{4d}{K_{eq} \times A} = \frac{d}{3KA} + \frac{3d}{KA}$$

$$\Rightarrow \frac{4}{K_{eq}} = \frac{1}{3K} + \frac{3}{K} \Rightarrow \frac{4}{K_{eq}} = \frac{1}{K} \left[\frac{10}{3} \right]$$

$$\Rightarrow K_{eq} = \frac{4 \times 3 \times K}{10} \Rightarrow K_{eq} = \frac{6K}{5}$$

(7) Answer : (4)

Solution:

The average velocity of gas molecules is zero due to motion of the molecule in random direction.

(8) Answer : (3)

Solution:



$$E_2 - E_1 = h\nu$$

$$\Rightarrow \nu = \frac{E_2 - E_1}{h}$$

(9) Answer : (3)

Solution:

No. of photons emitted per second $n = \frac{P}{\frac{hc}{\lambda}}$

$$n = \frac{200 \times 450 \times 10^{-9}}{6 \times 10^{-34} \times 3 \times 10^8}$$

$$n = \frac{200 \times 450 \times 10^{-9+34-8}}{6 \times 3}$$

$$= 5 \times 10^{20}$$

(10) Answer : (4)

Solution:

Energy is released when nuclei fuse to form more stable nuclei or breaks to form more stable nuclei.

$\therefore$ (i), (ii) and (iii) only

(11) Answer : (3)

Solution:

Barrier potential offers opposition to majority carriers in both regions and support motion of minority carriers.

(12) Answer : (3)

Solution:

$$i_L = \frac{12}{10^3} = 12 \text{ mA}$$

$$i_{Rs} = \frac{8}{250} = 32 \text{ mA}$$

$$i_z = i_{Rs} - i_L$$

$$= 32 - 12$$

$$= 20 \text{ mA}$$

(13) Answer : (3)

Hint:

Hint: Amplitude halves every 20 oscillations.

Solution:

Half life of this motion is corresponding to 20 oscillations.

$$A = \frac{A_0}{2^n}$$

Corresponding to 60 oscillations, there would be 3 half lives.

Here $n = 3$

$$\therefore A = \frac{24}{2^3}$$

$$= \frac{24}{8} = 3$$

(14) Answer : (2)

Solution:

$$\text{Torque } \tau = \frac{dL}{dt} \Rightarrow dL = \tau dt$$

$$\int_4^L dL = \int_0^4 4t dt$$

$$\Rightarrow L - 4 = 4 \times \frac{4^2}{2} = 32$$

$$\Rightarrow L = 36 \text{ kg m}^2 \text{ s}^{-1}$$

(15) Answer : (2)

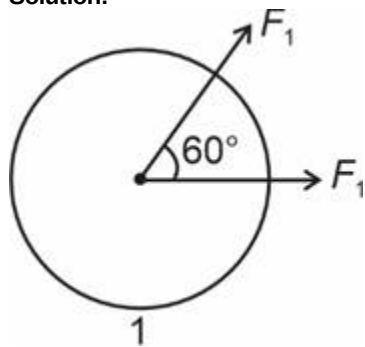
Solution:

$$I_O = I_{cm} + m \left(\frac{L}{\sqrt{12}} \right)^2 = \frac{ml^2}{12} + \frac{ml^2}{12} = \frac{ml^2}{6}$$

$$\text{K.E.} = \frac{1}{2} I_O \omega^2 = \frac{1}{2} \times \frac{ml^2}{6} \times (2\pi n)^2 = \frac{ml^2 \pi^2 n^2}{3}$$

(16) Answer : (4)

Solution:



$$F_1 = \frac{GMM}{(2R)^2} = \frac{G \left(\frac{4}{3} \pi R^3 \rho \right)^2}{4R^2} \propto R^4$$

$$F = 2F_1 \cos \left(\frac{60^\circ}{2} \right) = 2F_1 \cos 30^\circ = \sqrt{3}F_1 \Rightarrow F \propto R^4$$

(17) Answer : (2)

Solution:

$$\vec{I} = \frac{GM}{r^2} (-\hat{r}) \text{ for } r \geq R, \text{ where } R = \text{radius of solid sphere}$$

$$\vec{I} = \frac{GM}{R^3} (-\hat{r}) \text{ } r < R$$

(18) Answer : (4)

Solution:

$$\frac{1}{f_{\text{air}}} = \left(\frac{\mu_{\text{lens}}}{\mu_{\text{air}}} - 1 \right) \frac{2}{R} = \left(\frac{3}{2} - 1 \right) \frac{2}{R} = \frac{1}{R} = P_{\text{air}}$$

$$\frac{1}{f_{\text{water}}} = \left(\frac{\mu_{\text{lens}}}{\mu_{\text{water}}} - 1 \right) \frac{2}{R} = \left(\frac{3}{\frac{4}{3}} - 1 \right) \frac{2}{R} = \left(\frac{9}{8} - 1 \right) \frac{2}{R} = \frac{1}{4R}$$

$$f_{\text{water}} = 4f_{\text{air}}$$

(19) Answer : (3)

Solution:

$$\text{Lens formula, } \frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{v}{u} = \frac{f}{f+u} + 1$$

$$\Rightarrow \text{Magnification, } m = \frac{v}{u} = \frac{f}{f+u}$$

$$\text{Virtual image} \Rightarrow m_1 = \frac{f}{f-30} \text{ and real image}$$

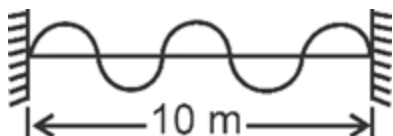
$$\Rightarrow m_2 = \frac{f}{f-50}$$

$$m_1 = -m_2 \Rightarrow f - 30 = -(f - 50)$$

$$\Rightarrow 2f = 80 \Rightarrow f = 40 \text{ cm}$$

(20) Answer : (2)

Solution:



$$\frac{5\lambda}{2} = 10, \lambda = 4 \text{ m}$$

$$v = \lambda \times f$$

$$\frac{20}{4} = f = 5 \text{ Hz}$$

(21) Answer : (3)

Solution:

In SHM acceleration (a) = $-\omega^2 A \sin \omega t$

$$a = -\omega^2 x$$

$\therefore a = -2x$ represents SHM

(22) Answer : (3)

Solution:

$$B = -\frac{PV}{\Delta V}$$

$$V = \frac{4}{3}\pi r^3, \Delta V = \frac{4}{3}\pi(3r^2)\Delta r$$

$$-\frac{V}{\Delta V} = -\frac{r}{3\Delta r}$$

$$B = -\frac{Pr}{3\Delta r} \therefore -\frac{\Delta r}{r} = \frac{P}{3B}$$

(23) Answer : (1)

Solution:

$$\vec{v} = \vec{E} \times \vec{B}$$

$$\hat{k} = \hat{i} \times \hat{j}$$

$$\therefore \vec{E} = E(t)\hat{i}, \vec{B} = B(t)\hat{j}$$

(24) Answer : (2)

Solution:

- Concept of secondary wavelets is given by Huygen's principle.
- Law of Malus gives intensity of light passing through analyser.
- All bright fringes are of equal width in YDSE.
- In diffraction, intensity of secondary bright fringes keeps on decreasing.

(25) Answer : (2)

Solution:

If the number of observations is made x times then the random error is reduced to $\left(\frac{1}{x}\right)$ times.

(26) Answer : (2)

Solution:

$$V = \frac{4}{3}\pi r^3$$

$$\left(\frac{\Delta V}{V} \times 100\right) = 3 \left(\frac{\Delta r}{r} \times 100\right)$$

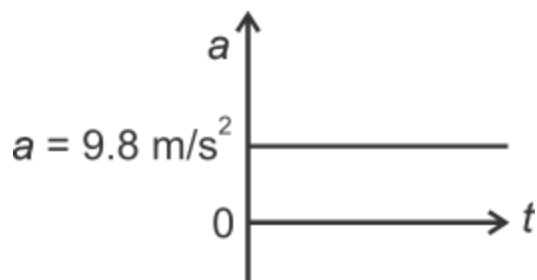
$$\text{Area} = 4\pi r^2$$

$$\left(\frac{\Delta A}{A} \times 100\right) = 2 \left(\frac{\Delta r}{r} \times 100\right)$$

$$\text{Hence, } \left(\frac{\Delta V}{V} \times 100\right) > \left(\frac{\Delta A}{A} \times 100\right)$$

(27) Answer : (3)

Solution:



For free fall, particle is under uniform acceleration.

(28) Answer : (3)

Solution:

$$a = \frac{\Delta v}{\Delta t}$$

In straight line motion, if particle is speeding up then acceleration will be in direction of velocity.

(29) Answer : (2)

Solution:

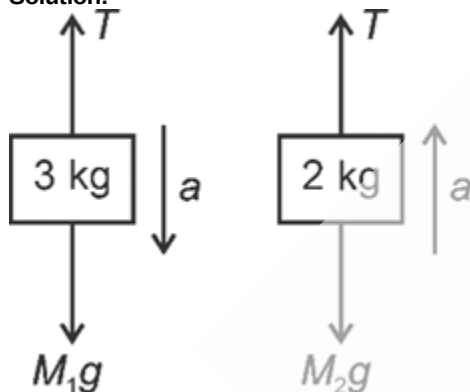
For a ideal spring, $KL = \text{Constant}$

$$K_1 = \frac{KL}{\frac{L}{3}} = 3K$$

$$K_2 = \frac{3K}{2}$$

(30) Answer : (2)

Solution:



$$M_1g - T = M_1a$$

$$T - M_2g = M_2a$$

Solving these, we get

$$a = 2 \text{ m/s}^2 \text{ and } T = 24 \text{ N}$$

(31) Answer : (2)

Solution:

$$K.E = \frac{1}{2}mv^2 \times \frac{m}{m}$$

$$K.E = \frac{1}{2} \frac{m^2v^2}{m}$$

$$E = \frac{P^2}{2m}$$

$$\sqrt{E} = \frac{P}{\sqrt{2m}}$$

$$\sqrt{E} \propto P$$

(32) Answer : (2)

Solution:

$$m_1 = 0.1 \text{ kg}$$

$$u_2 = 0$$

$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) u_1 + \frac{2m_2u_2}{m_1 + m_2}$$

$$-\frac{u}{3} = \left(\frac{0.1 - m_2}{0.1 + m_2} \right) u \Rightarrow m_2 = 0.2 \text{ kg}$$

(33) Answer : (1)

Solution:

$$\phi = 2t^2 + 4t + 4$$

$$|\varepsilon| = \frac{d\phi}{dt}$$

$$|\varepsilon|_{t=2} = 4t + 4$$

$$|\varepsilon|_{t=2} = 12 \text{ V}$$

$$I = \frac{V}{R}$$

$$= \frac{12}{4} = 3 \text{ A}$$

(34) Answer : (2)

Solution:

$$L \propto n^2$$

$$\frac{L}{L_1} = \frac{n^2}{(2n)^2}$$

$$L_1 = 4 L$$

(35) Answer : (3)

Solution:

Since the coils are at right angles to each other, net magnetic field will be:

$$B = \sqrt{B_1^2 + B_2^2}$$

$$= \sqrt{\left(\frac{3\mu_0 I}{2R}\right)^2 + \left(\frac{4\mu_0 I}{2R}\right)^2}$$

$$= \left(\frac{5\mu_0 I}{2R}\right)$$

(36) Answer : (4)

Solution:

$$\vec{F} = q(\vec{v} \times \vec{B})$$

$\vec{F}$ and $\vec{v}$ are perpendicular to each other.

$$\therefore \vec{F} \perp \vec{B} \text{ and } \vec{F} \perp \vec{v}$$

(37) Answer : (3)

Solution:

$$\cos \phi = \frac{V_R}{\sqrt{V_R^2 + (V_L - V_C)^2}}$$

$$= \frac{80}{\sqrt{80^2 + (60)^2}}$$

$$= \frac{80}{100}$$

$$= \frac{4}{5}$$

$$\phi = 37^\circ$$

(38) Answer : (3)

Solution:

$$X = \omega L$$

$$X = 2\pi fL$$

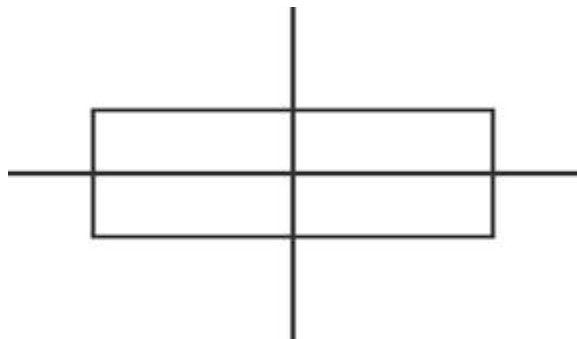
$$X' = 2\pi 2f \times 2 L$$

$$= 8\pi fL$$

$$= 4X$$

(39) Answer : (3)

Solution:



Pole strength of each part will be $\frac{m}{2}$

(40) Answer : (4)

Solution:

Both the spheres will be at same potential.

$$V_1 = V_2 \Rightarrow \frac{Kq_1}{a} = \frac{Kq_2}{b} \dots(i)$$

$$q_1 = \sigma_1 \times 4\pi a^2, \quad q_2 = \sigma_2 \times 4\pi b^2$$

$$\frac{K\sigma_1 \times 4\pi a^2}{a} = \frac{K\sigma_2 \times 4\pi b^2}{b}$$

$$\frac{\sigma_1}{\sigma_2} = \frac{b}{a}$$

(41) Answer : (1)

Solution:

In an isolated system, net charge remains conserved, hence equal amount of positive and negative charge is produced or destroyed.

(42) Answer : (3)

Solution:

We will need four identical cubes to enclose the charge. Due to symmetry, flux through each cube will be equal.

$$\phi_{\text{net}} = \oint \vec{B} \cdot d\vec{A} = \frac{q_{\text{in}}}{\epsilon_0}$$

$$\phi_{\text{net}} = \frac{4}{\epsilon_0} \dots(i)$$

Flux through one cube:

$$\phi = \frac{\phi_{\text{net}}}{4} = \frac{4}{\epsilon_0} \times \frac{1}{4} = \frac{1}{\epsilon_0}$$

(43) Answer : (2)

Solution:

System without dielectric:

$$C_{eq} = \frac{C}{2}, \quad Q_0 = \frac{CV}{2} \dots(i)$$

System with dielectric:

$$C'_{eq} = \frac{C \times 2C}{C + 2C} = \frac{2C}{3}$$

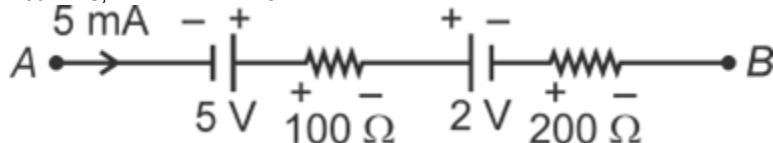
$$Q' = \frac{2C}{3} \times V \dots(ii)$$

$$\frac{Q_0}{Q'} = \frac{\frac{CV}{2}}{\frac{2CV}{3}} = \frac{3}{4} \Rightarrow Q' = \frac{4Q_0}{3}$$

(44) Answer : (4)

Solution:

At $t = 2$ s, $i = 2 \times 2 + 1 = 5$ mA



$$V_A + 5 - (100 \times 5 \times 10^{-3}) - 2 - (200 \times 5 \times 10^{-3}) = V_B$$

$$V_A + 3 - 0.5 - 1 = V_B$$

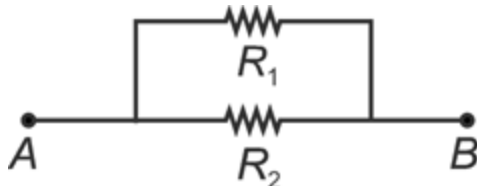
$$V_A - V_B = -1.5 \text{ V}$$

(45) Answer : (1)

Solution:

$$R_1 = R_0 \times 2r = 2R_0r \dots (i)$$

$$R_2 = R_0 \times \pi r = \pi R_0r \dots (ii)$$

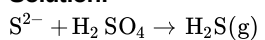


$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2} = \frac{2R_0r \times \pi R_0r}{2R_0r + \pi R_0r} = \frac{2\pi R_0r}{2 + \pi}$$

CHEMISTRY

(46) Answer : (4)

Solution:

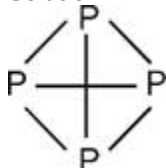


Rotten egg

like smell

(47) Answer : (3)

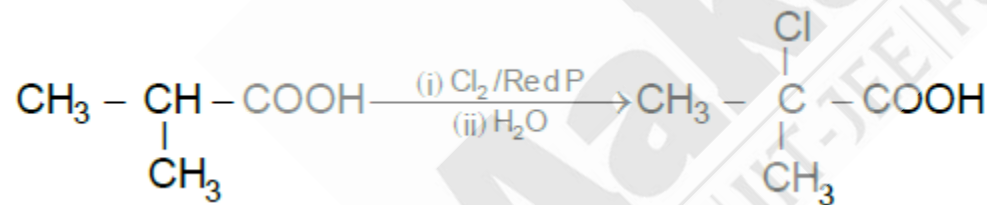
Solution:



P-P-P bond angle is 60° .

(48) Answer : (2)

Solution:



(49) Answer : (4)

Solution:

Ketones containing α -hydrogen atoms do not undergo Cannizzaro's reaction.

(50) Answer : (2)

Solution:

It is a first order reaction

$$t_{1/2} = \frac{0.693}{k}$$

$$= \frac{0.693}{1.386 \times 10^{-3}} = 500 \text{ s}$$

(51) Answer : (3)

Solution:

Catalyst catalyses spontaneous reactions only by decreasing activation energy.

(52) Answer : (2)

Solution:

Increase in concentration of reactants, increase in pressure and decrease in temperature will favour the production of NH_3 .

(53) Answer : (3)

Solution:

$$K_P = K_C(RT)^{\Delta ng}$$

for $K_P = K_C$, $\Delta n_g = 0$

and in $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

$\Delta n_g = 0$

so, $K_P = K_C$

(54) Answer : (1)

Solution:

NO , CO , N_2O are neutral oxides, while As_2O_3 , Al_2O_3 are amphoteric oxide. NO_2 is acidic oxide.

(55) Answer : (2)

Solution:

Group number = $(ns + (n-1)d)$ electron

$= 2 + 7 = 9$

(56) Answer : (2)

Solution:

$Fe^{3+}(aq) + 3e^- \rightarrow Fe(s)$ $E^\circ = x \text{ volt}$ $\Delta G_1^\circ = -3Fx$

$Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}(aq)$ $E^\circ = y \text{ volt}$ $\Delta G_2^\circ = -Fy$

(i) – (ii) will give

$Fe^{2+}(aq) + 2e^- \rightarrow Fe(s)$ ΔG_3°

(z is the reduction potential of above half cell reaction)

$\Delta G_3^\circ = -2Fz = -3Fx + Fy$

$z = \frac{3x - y}{2} \text{ volts}$

(57) Answer : (3)

Solution:

Ion	$\lambda^\circ (\text{Scm}^2 \text{mol}^{-1})$
OH^-	199.1
SO_4^{2-}	160.0
Br^-	78.1
Cl^-	76.3

(58) Answer : (2)

Solution:

$\Delta T_f = k_f \times m$

$0.5 = \frac{5.12 \times 2 \times 1000}{M \times 100}$

$M = \frac{5.12 \times 2 \times 1000}{0.5 \times 100}$

$= 204.8 \text{ g/mol}$

(59) Answer : (4)

Solution:

In case of positive deviation from Raoult's law, A-B interactions are weaker than those between A-A or B-B interactions.

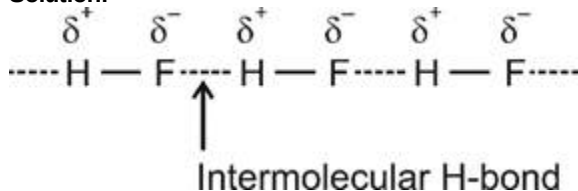
(60) Answer : (1)

Solution:

The electrons are removed from π^*2p_x or π^*2p_y molecular orbitals.

(61) Answer : (4)

Solution:



(62) Answer : (3)

Solution:

Molecule	Bond angle
	$\rightarrow 109.5^\circ$

CH ₄		
H ₂ O	→	104.5°
NH ₃	→	107°

(63) Answer : (2)

Solution:

Ethane-1,2-diamine is bidentate ligand

Coordination number of cobalt in $[\text{CoCl}_2(\text{en})_2]^+$ is six.

(64) Answer : (4)

Solution:

The crystal field splitting, Δ_0 , depends upon the field produced by ligand and charge on the metal ion.

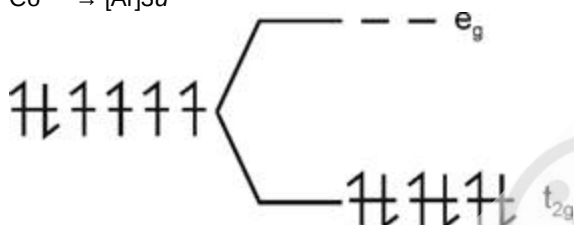
In $[\text{Fe}(\text{CN})_6]^{3-}$

$\text{Fe}^{3+} \rightarrow [\text{Ar}]3d^5$

$\text{CFSE} = -0.4 \times 5\Delta_0 = -2.0\Delta_0$

In $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$

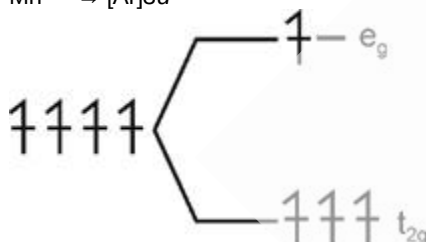
$\text{Co}^{3+} \rightarrow [\text{Ar}]3d^6$



$\text{CFSE} = -0.4 \times 6\Delta_0 = -2.4\Delta_0$

In $[\text{MnCl}_6]^{3-}$

$\text{Mn}^{3+} \rightarrow [\text{Ar}]3d^4$



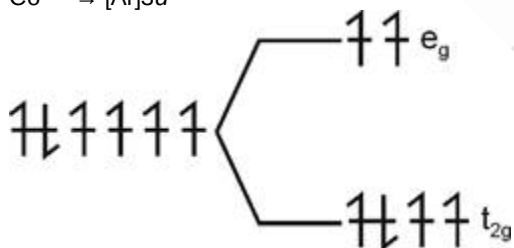
$\text{CFSE} = -0.4\Delta_0 \times 3 + 0.6\Delta_0 \times 1$

$= -1.2\Delta_0 + 0.6\Delta_0$

$= -0.6\Delta_0$

In $[\text{CoF}_6]^{3-}$

$\text{Co}^{3+} \rightarrow [\text{Ar}]3d^6$



$\text{CFSE} = -0.4\Delta_0 \times 4 + 0.6\Delta_0 \times 2$

$= -1.6\Delta_0 + 1.2\Delta_0$

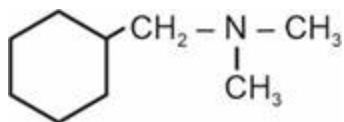
$= -0.4\Delta_0$

(65) Answer : (3)

Solution:

Benzenesulphonyl chloride ($\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$), is known as Hinsberg's reagent

Tertiary amines do not react with benzene sulphonyl chloride.

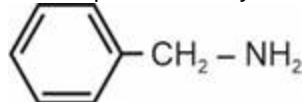


is 3° amine so it does not react with Hinsberg's reagent.

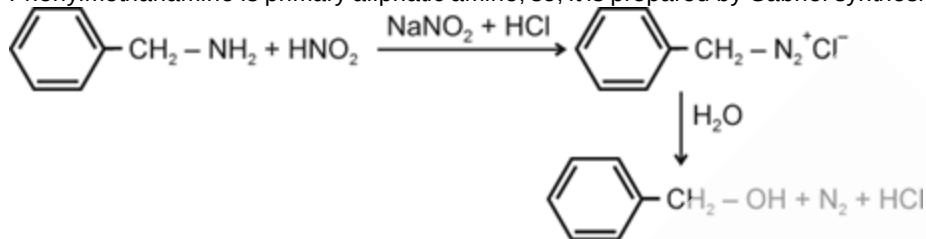
(66) Answer : (4)

Solution:

Gabriel phthalimide synthesis is used for the preparation of aliphatic primary amines.



Phenylmethanamine is primary aliphatic amine, so, it is prepared by Gabriel synthesis.

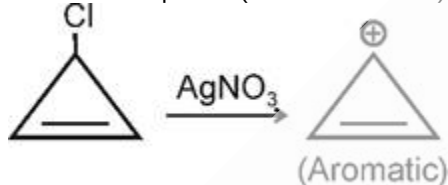


(67) Answer : (1)

Solution:



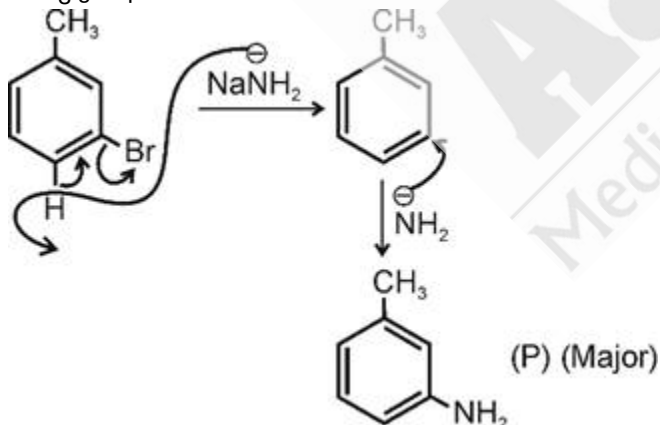
it is aromatic species ($2\pi e^-$ delocalised)



(68) Answer : (2)

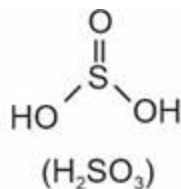
Solution:

$-\text{CH}_3$ group has +I effect.

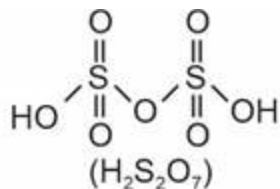


(69) Answer : (4)

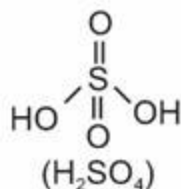
Solution:



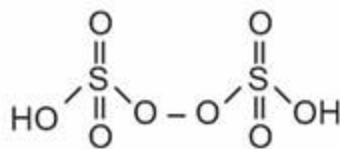
Sulphurous acid



Pyrosulphuric acid



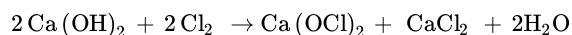
Sulphuric acid



Peroxodisulphuric acid

(70) Answer : (3)

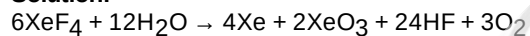
Solution:



Slaked lime

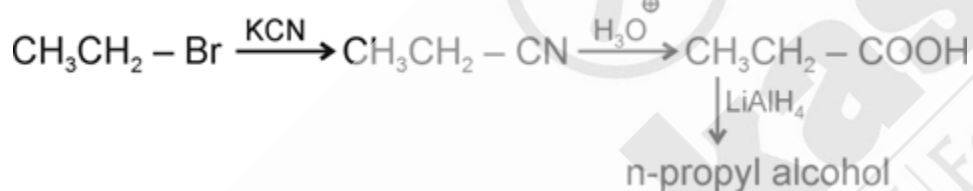
(71) Answer : (4)

Solution:



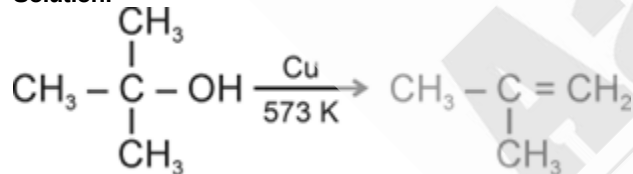
(72) Answer : (2)

Solution:



(73) Answer : (3)

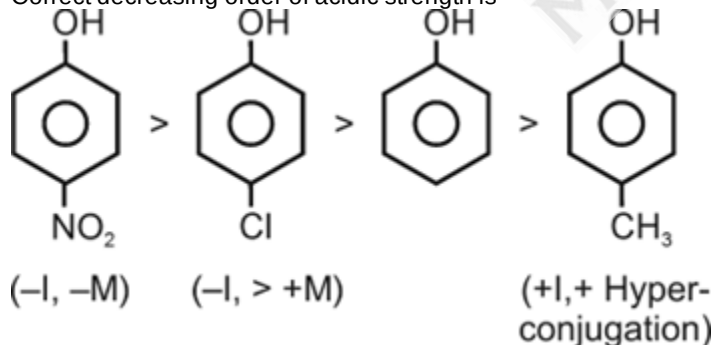
Solution:



(74) Answer : (1)

Solution:

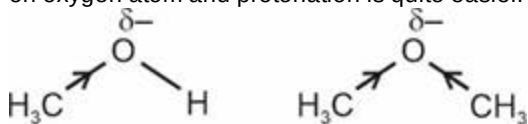
Correct decreasing order of acidic strength is



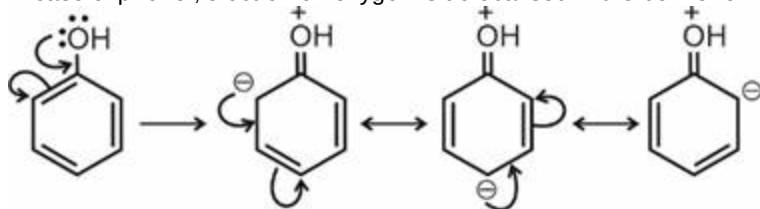
(75) Answer : (1)

Solution:

–CH₃ group in methanol (CH₃OH) and methoxymethane (CH₃ – O – CH₃) has +I effect that increases the electron density on oxygen atom and protonation is quite easier.



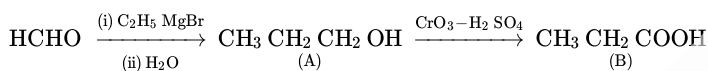
In case of phenol, electron on oxygen is delocalised in the benzene ring.



So, phenol is most difficult to protonate.

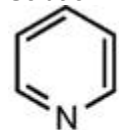
(76) Answer : (1)

Solution:



(77) Answer : (4)

Solution:



Pyridine



Furan



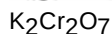
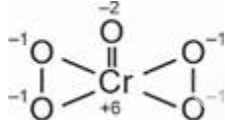
Thiophene



Tropone

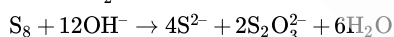
(78) Answer : (4)

Solution:



$$\Rightarrow 2(+1) + 2x + (-2) \times 7 = 0$$

$$\therefore x = \frac{12}{2} = +6$$



(79) Answer : (3)

Solution:

For group 13 elements, +1 oxidation is stable for heavier members of the group due to inert pair effect. The order of stability of +1 oxidation state is in the order of Al < Ga < In < Tl

(80) Answer : (1)

Solution:

Melting point of α-form of glucose is 419 K while that of β-form of glucose is 423 K.

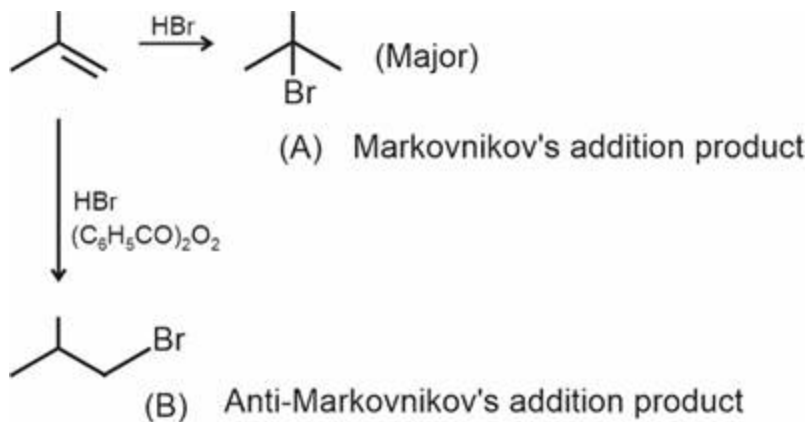
(81) Answer : (2)

Solution:

Maximum number of electrons in a subshell(l) = 2(2l + 1)

(82) Answer : (1)

Solution:



(83) Answer : (1)

Solution:

During denaturation of protein, secondary and tertiary structures are destroyed but primary structure remains intact.

(84) Answer : (4)

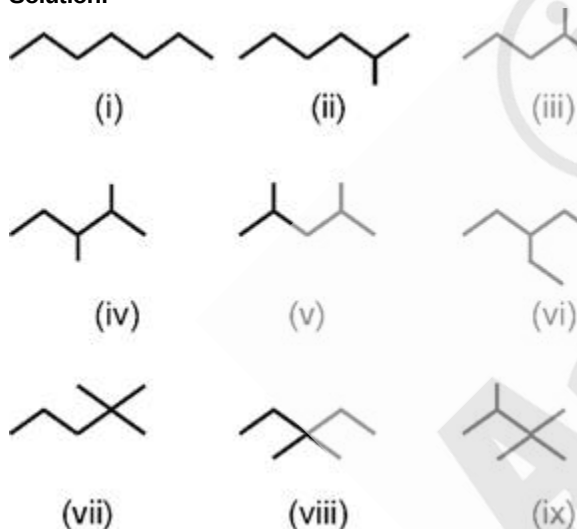
Solution:

$$\frac{e}{m} \text{ of electron} = \frac{1.602 \times 10^{-19}}{9.109 \times 10^{-31}} = 0.1758 \times 10^{12} \text{ C kg}^{-1}$$

$$= 1.758 \times 10^{11} \text{ C kg}^{-1}$$

(85) Answer : (3)

Solution:



(86) Answer : (1)

Solution:

98 g conc. H_2SO_4 is present in 100 g solution
 moles of $\text{H}_2\text{SO}_4 = 1$

$$\text{volume of solution} = \frac{100}{1.84} \text{ mL}$$

Molarity of conc. H_2SO_4 solution

$$= \frac{98 \times 1000}{98 \times \frac{100}{1.84}} = 18.4$$

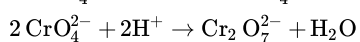
$$M_1 V_1 = M_2 V_2$$

$$18.4 \times V_1 = 2.5 \times 0.5$$

$$V_1 = \frac{1.25}{18.4} = 0.068 \text{ L} \approx 68 \text{ mL}$$

(87) Answer : (1)

Solution:



(88) Answer : (2)

Solution:

Ions	Spin magnetic moment (BM)
Fe^{3+}	5.92
Cu^{+}	0
Co^{3+}	4.89
V^{2+}	3.87

$$\mu = \sqrt{n(n+2)} \text{ B.M}$$

μ = spin magnetic moment

n = number of unpaired electrons.

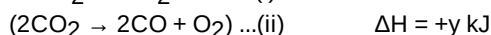
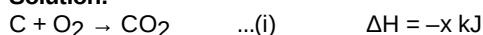
(89) Answer : (3)

Solution:

Ba^{2+} , Sr^{2+} and Ca^{2+} are group-V cations. The group reagent is $(\text{NH}_4)_2\text{CO}_3$ in presence of NH_4OH .

(90) Answer : (4)

Solution:



$$(\text{i}) + \left(\frac{\text{ii}}{2}\right)$$

$$\text{i.e., } \frac{y-2x}{2} = \Delta H$$

BIOLOGY

(91) Answer : (2)

Solution:

Water-containing cavities are present within the vascular bundles of monocot stem.

(92) Answer : (3)

Solution:

Vessel – Vessel members are interconnected through perforation in their common walls.

Phloem parenchyma – Made up of elongated, tapering cylindrical cells which have dense cytoplasm and nucleus.

Sclereids – Found in pulp of pear and sapota.

Collenchyma – Occurs in layers below epidermis in most of the dicots.

(93) Answer : (2)

Solution:

One codon has three bases, and it codes for 1 amino acid, so deletion of base at 601 position will lead to alteration in 22 codons.

(94) Answer : (4)

Solution:

DNA-dependent DNA polymerases catalyse the polymerisation of deoxynucleotides.

(95) Answer : (4)

Solution:

Structural gene is polycistronic in prokaryotes (eg. *E. coli*). Genes are split in eukaryotes.

(96) Answer : (2)

Solution:



A translational unit in mRNA is the sequence of RNA that is flanked by the start codon (AUG) and stop codon and codes for polypeptide.

(97) Answer : (2)

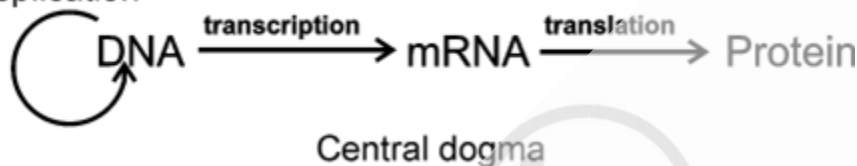
Solution:

The given tissue is collenchyma and found to be absent in monocots. Parenchyma forms major component within organs. At the corners of these cells, deposition of cellulose, hemicellulose and pectin is found.

(98) Answer : (3)

Solution:

replication



(99) Answer : (2)

Solution:

Some bacteria have a loose sheath called slime layer, which protects the cell from the loss of water and nutrients.

(100) Answer : (3)

Solution:

"9 + 0" organization of microtubules is observed in the basal body of cilia or flagella in eukaryotes.

(101) Answer : (3)

Solution:

Lysosomes contain hydrolytic enzymes for digesting macromolecules. These enzymes are active in acidic conditions.

(102) Answer : (2)

Solution:

Prophase is the first stage of mitosis, where there is condensation of chromatin material and nuclear envelope breaks down and nucleolus disappear.

(103) Answer : (3)

Solution:

Cis face of Golgi body is the forming face.

(104) Answer : (3)

Solution:

The correct sequence of events is as follows:

- Pachytene (Exchange of genetic material)
- Diplotene (Chiasmata formation)
- Diakinesis (Nucleolus disappears and the nuclear envelope disintegrates)
- Telophase I (Nucleolus and nuclear membrane reappear)
- Anaphase II (Splitting of centromere holding the two chromatids)

(105) Answer : (1)

Solution:

Taxonomic categories showing hierarchial arrangement in ascending order is
Species → Genus → Family → Order → Class → Phylum → Kingdom

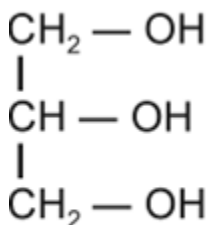
(106) Answer : (1)

Solution:

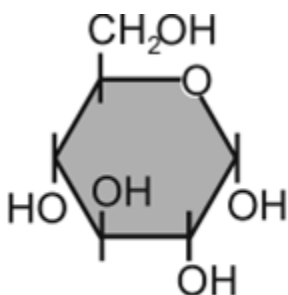
In the dense regular connective tissues, the collagen fibres are present in rows between many parallel bundles of fibres. Tendons which attach skeletal muscles to bones and ligaments which attach one bone to another are examples of this tissue.

(107) Answer : (3)

Solution:



Glycerol



$\text{C}_6\text{H}_{12}\text{O}_6$ (Glucose)

(108) Answer : (2)

Solution:

Certain amino acids are essential for our health and they have to be supplied through our diet. Hence, dietary proteins are the source of essential amino acids. The non-essential amino acids are those which our body can make. Paper made from plant pulp is cellulosic in nature.

(109) Answer : (2)

Solution:

Lipids are indeed small molecular weight compounds and are present not only as such but also arranged into structures like cell membrane and other membranes. When we grind a tissue, cell membrane and other membranes are broken into pieces and form vesicles which are not water soluble. Therefore, these membrane fragments in the form of vesicles get separated along with the acid-insoluble pool and hence remain in the macromolecular fraction. Lipids are not strictly macromolecules.

(110) Answer : (3)

Solution:

Exoskeleton of arthropods have a complex polysaccharide called chitin whose monomeric unit is N-acetyl glucosamine.

(111) Answer : (4)

Solution:

In all connective tissues (except blood), the cells secrete fibres of structural proteins called collagen or elastin. The fibres provide strength, elasticity and flexibility to the tissue.

(112) Answer : (4)

Solution:

In IUT, embryo with more than 8 blastomeres collected from healthy donor is artificially introduced either into the vagina or into the uterus.

(113) Answer : (3)

Solution:

The enlarged end of penis is called glans penis which is covered by a loose fold of skin called foreskin.

(114) Answer : (2)

Solution:

Sertoli cells provide nutrition to germ cells. Leydig cells/interstitial cells secrete hormones which are called androgens. Male germ cells are present in germinal epithelium of seminiferous tubules of testis.

(115) Answer : (1)

Solution:

Frogs have the ability to change their body colour to hide themselves from their enemies. This protective coloration is called mimicry. *Periplaneta* and humans do not show mimicry.

(116) Answer : (1)

Solution:

Halophiles are archaebacteria that can survive in extremely salty areas.

(117) Answer : (3)

Solution:

Mad cow disease in cattle is caused by prions which consist of abnormally folded protein.

(118) Answer : (3)

Solution:

Root hair is the outer extension of epidermal cells. It increases the surface area for absorption of water and minerals from soil.

(119) Answer : (2)**Solution:**

Types of placentation	Example
Marginal	Pea
Axile	Lemon
Parietal	<i>Argemone</i>
Free central	<i>Dianthus</i>

(120) Answer : (3)**Solution:**

In mustard, female reproductive part (ovary) occupies the highest position while the other parts are situated below it. Such flowers are called hypogynous flowers.

(121) Answer : (4)**Solution:**

Mycoplasma is the smallest wall-less moneran, which is mostly parasitic.

(122) Answer : (1)**Solution:**

Biomass of fish is larger as fishes are larger in size with longer life span and a number of generations of zooplanktons can be consumed by fishes.

Sum total of biomass of benthic animals for algae exceeds the producers and consumers in aquatic ecosystem.

(123) Answer : (3)**Solution:**

Jan Ingenhousz showed that sunlight is essential to the plant process that somehow purifies the air fouled by burning candles or breathing animals.

(124) Answer : (2)**Solution:**

Chlorophyll *b* pigment shows the highest peak in the absorption spectrum between 400-500 nm wavelength of light.

(125) Answer : (2)**Solution:**

Less than 7% of energy in glucose is released during both alcoholic and lactic acid fermentation.

(126) Answer : (4)**Solution:**

Under anaerobic conditions, the rate of decomposition is slow.

(127) Answer : (3)**Solution:**

	Column I		Column II
(a)	Complex I	(i)	NADH Dehydrogenase
(b)	Complex II	(ii)	Succinate dehydrogenase
(c)	Complex III	(iii)	Cytochrome <i>bc</i> ₁ complex
(d)	Complex IV	(iv)	Cytochrome <i>c</i> oxidase

(128) Answer : (4)**Solution:**

In gymnosperms, the microspores develop into a male gametophytic generation which is highly reduced and is confined to only a limited number of cells. This reduced gametophyte is called as pollen grain.

(129) Answer : (4)**Solution:**

Dryopteris is a pteridophyte, in which the spores germinate to give rise to inconspicuous, small but multicellular, free-living, mostly photosynthetic thalloid gametophytes called prothallus.

(130) Answer : (2)**Solution:**

Toddy – *Caryota urens*

Roquefort cheese – *Penicillium roquefortii*

Baker's yeast – *Saccharomyces*

Curd – *Lactobacillus*

(131) Answer : (1)

Solution:

Ovaries are the primary sex organ in human females which produce ova. They also produce a number of steroidal sex hormones such as estrogens and progesterone. Vagina, cervix, uterus and oviducts are secondary sex organs in females.

(132) Answer : (4)

Solution:

Head of cockroach lies anteriorly at right angles to the longitudinal axis of the body. Head is formed by the fusion of 6 segments in adults. Head capsule bears a pair of compound eyes.

(133) Answer : (2)

Solution:

Chorionic villi are finger-like projections of trophoblast which are surrounded by maternal tissue. Morula consists of 8-16 blastomeres. The blastocyst consists of outer layer called trophoblast and the inner cell mass.

(134) Answer : (2)

Solution:

Genetically engineered plants have enhanced nutritional value. For example-Golden rice contains β -carotene, precursor of vitamin-A.

Pomato was not successful at commercial level. Retroviruses are used for gene transfer in animals.

(135) Answer : (3)

Solution:

Gene therapy is a collection of methods that allows correction of a gene defect that has been diagnosed in a person.

(136) Answer : (3)

Solution:

Pheretima – Terrestrial organism

Anopheles – Internal fertilisation

Spongilla – Water canal system

Water vascular system is present in echinoderms.

(137) Answer : (2)

Solution:

Aquatic annelids like *Nereis* possess lateral appendages, parapodia, which help in swimming.

(138) Answer : (4)

Solution:

In 1983, Eli Lilly produced humulin for the first time. Prior to that, human insulin was not available so insulin extracted from pancreas of slaughtered cattle and pigs was used.

(139) Answer : (4)

Solution:

Reptiles, amphibians and fishes are cold-blooded i.e., poikilotherms. Hence, all the mentioned organisms are poikilotherms.

(140) Answer : (4)

Solution:

By the time of 500 mya, invertebrates were formed and active.

(141) Answer : (2)

Solution:

Sauropsids were the predecessors of thecodonts.

(142) Answer : (2)

Solution:

Horsetails, ginkgos, ferns and conifers have evolved directly from Psilophyton. Seed ferns evolved from progymnosperms while cycads evolved from seed ferns.

(143) Answer : (4)

Solution:

The CNS passes on motor messages to initiate the contraction of smooth muscles of the urinary bladder and simultaneous relaxation of the urethral sphincter causing the release of urine.

(144) Answer : (3)

Solution:

Modern *Homo sapiens* arose during ice age between 75,000-10,000 years ago. The Neanderthal man lived in near east and central Asia between 1,00,000-40,000 years back.

(145) Answer : (2)

Solution:

The collecting duct allows passage of small amounts of urea into the medullary interstitium to keep up the osmolarity. It also plays a role in the maintenance of pH and ionic balance of the blood by selective secretion of H^+ and K^+ .

(146) Answer : (4)

Solution:

Rheumatoid arthritis and Myasthenia gravis are auto-immune disorders. Muscular dystrophy is a genetic disorder. If our body becomes incapable of distinguishing between foreign organisms from self cells, it can lead to auto-immune disorders.

(147) Answer : (4)

Solution:

The Leydig cells or interstitial cells produce a group of hormones called androgens, mainly testosterone. They regulate development, maturation and functions of the male accessory sex organs like epididymis, vas deferens, seminal vesicles, prostate gland, etc.

These hormones produce anabolic(synthetic) effects on protein and carbohydrate metabolism.

(148) Answer : (4)

Solution:

Parathyroid glands are 4 in number in humans. They are present on the dorsal side of thyroid gland, one pair each in the two lobes of thyroid gland.

Thymus is located between lungs behind sternum on the ventral side of aorta.

Thyroid gland is composed of two lobes which are located on the either side of trachea.

(149) Answer : (2)

Solution:

Many fungi belonging to the genera *Microsporum*, *Trichophyton* and *Epidermophyton* are responsible for ringworm.

Heat and moisture help these fungi to grow, especially in skinfolds such as those in the groin or between the toes.

(150) Answer : (2)

Solution:

Help from parents and peers should be sought immediately so that they can guide appropriately. Help may even be sought from close and trusted friends.

(151) Answer : (1)

Solution:

Increase in H^+ concentration, higher temperature and high pCO_2 shifts the oxygen -dissociation curve to the right side.

(152) Answer : (2)

Solution:

Fibrous joint is present between the flat skull bones. The two halves of the pelvic girdle meet ventrally to form the pubic symphysis containing fibrous cartilage.

Pivot joint is a type of synovial joint which is present in between atlas and axis.

(153) Answer : (3)

Solution:

Emphysema leads to reduced respiratory surface. Lung fibrosis is seen in occupational respiratory disorders.

Lungs are not muscular structure.

(154) Answer : (1)

Solution:

Cardiac output = Heart rate $\times$ Stroke volume

5000 mL = H.R. $\times$ 50 mL

Heart rate = $5000/50 = 100$ beats/min

(155) Answer : (1)

Solution:

Myoglobin is a red-coloured oxygen storing pigment in muscles. In white muscle fibres, very less quantity of myoglobin is present. Haemoglobin is a red-coloured iron containing pigment present in RBCs which help in transport of gases.

(156) Answer : (4)

Solution:

Biological control methods adopted in agricultural pest control are based on the ability of the predator to keep prey population under control and be prudent in nature.

(157) Answer : (2)

Solution:

A stable community should not show too much variation in productivity from year to year.

(158) Answer : (2)

Solution:

Some examples of important recent extinctions include the Dodo (Mauritius), Quagga (Africa), Thylacine (Australia), Steller's sea cow (Russia).

(159) Answer : (4)**Solution:**

Gemmae are green, multicellular, asexual buds, which develop in small receptacles called gemma cups located on the thalli of *Marchantia*.

In *Equisetum* and *Selaginella*, sporophylls form distinct compact structures called strobili or cones, but only *Selaginella* has male and female strobili.

(160) Answer : (1)**Solution:**

The lady bird is useful to get rid of aphids and dragonflies are useful to get rid of mosquitoes.

(161) Answer : (3)**Solution:**

The monarch butterfly is highly distasteful to its predator (bird) because of a special chemical present in its body. It acquires the chemical during its caterpillar stage by feeding on a poisonous weed.

(162) Answer : (3)**Solution:**

Buttercup shows environmental heterophylly.

(163) Answer : (2)**Solution:**

Secondary growth is seen in dicot plants.

(164) Answer : (1)**Solution:**

Hybrid seeds have to be produced every year because if seeds from the hybrids are sown, the plants in the progeny will segregate and will not maintain hybrid characters.

(165) Answer : (2)**Solution:**

Orobanche is a parasitic plant species that contains a very large number of seeds.

(166) Answer : (3)**Solution:**

The early stages of embryogeny are similar in both monocots and dicots.

(167) Answer : (3)**Solution:**

Formula to calculate the number of types of gametes is 2^n , where 'n' is the number of heterozygous loci.
Hence, $2^4 = 16$

(168) Answer : (3)**Solution:**

Closely located genes assort together and distantly located genes, due to recombinations, assort independently.

(169) Answer : (1)**Solution:**

The given pedigree chart represents an autosomal recessive trait whereas myotonic dystrophy is an autosomal dominant trait.

(170) Answer : (3)**Solution:**

Polygenic traits are generally controlled by two or more genes.

(171) Answer : (3)**Solution:**

Myoglobin content is high in red muscle fibres. They contain plenty of mitochondria which can utilise the large amount of oxygen stored in them. White muscle fibres depend on anaerobic process for energy. They have high amount of sarcoplasmic reticulum.

(172) Answer : (4)**Solution:**

In a standard ECG, the T-wave represents the return of the ventricles from excited to normal state (repolarisation) and the end of the T-wave marks the end of systole. The P-wave represents the electrical excitation (or depolarisation) of the atria which leads to the contraction of both atria.

(173) Answer : (4)

Solution:

PCR involves 3 steps:

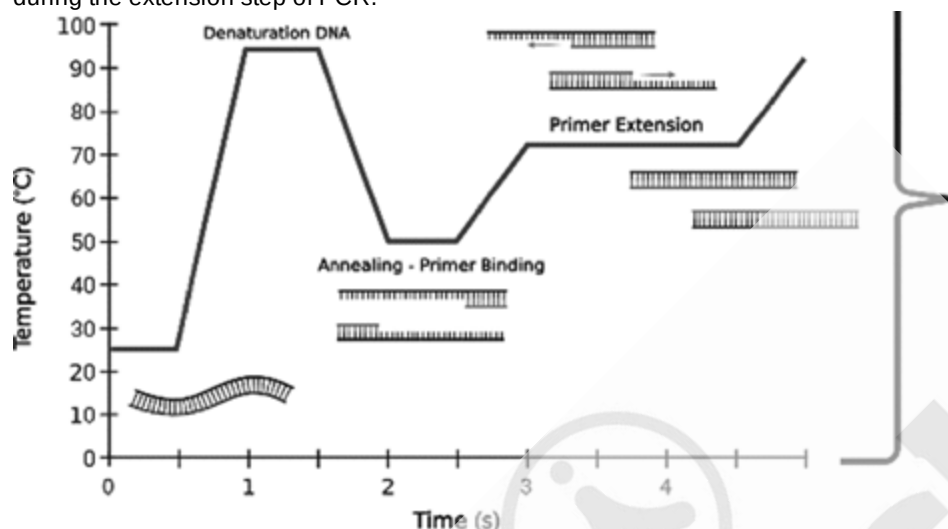
- (1) Denaturation
- (2) Annealing
- (3) Extension

Annealing of primers occurs at the 3'OH ends of the primers.

Extension occurs in the presence of heat-stable DNA polymerase.

Hydrogen bonds are broken during the denaturation step.

Primer anneals to a DNA template through hydrogen bonds during the PCR process. Phosphodiester bonds are formed during the extension step of PCR.



(174) Answer : (3)

Solution:

If a foreign gene is inserted at the *Pvu* I site of vector pBR322 followed by its insertion in host cell, then the recombinant transformants will be *amp*^S and *tet*^R and the non-recombinants will be *amp*^R and *tet*^R.

(175) Answer : (3)

Solution:

In the human brain, centres are located in the medulla oblongata that can regulate cardiac functions.

The autonomous nerve fibres connected to the medulla oblongata work to control heart rate with the release of neurotransmitters.

The dorsal portion of the midbrain consists mainly of four round swellings (lobes) called corpora quadrigemina.

(176) Answer : (3)

Solution:

When a neuron is not conducting any impulse, i.e., resting, the axonal membrane is comparatively more permeable to K^+ and nearly impermeable to Na^+ . While during depolarisation, the membrane becomes freely permeable to Na^+ , whereas during repolarisation, K^+ diffuses outside the membrane and restores the resting potential of the membrane at the site of excitation and the neuronal fibres once more become responsive to further stimulation.

(177) Answer : (1)

Solution:

In biolistics or gene gun method (suitable for plants), cells are bombarded with high velocity micro-particles of gold or tungsten coated with DNA.

(178) Answer : (3)

Solution:

Plasmids are autonomously replicating circular extra-chromosomal DNA.

(179) Answer : (1)

Solution:

The separated DNA fragments can be visualised only after staining the DNA with a compound known as ethidium bromide followed by exposure to UV radiation. Bright orange-coloured bands of DNA are visible.

Aniline blue is a dye used to stain tissues and cells.

Methylene blue is a salt used as a dye.

Bromophenol blue is used as a tracking dye during gel electrophoresis.

(180) Answer : (1)

Solution:

Progestogens alone or in combination with estrogen can be used by females as injection or implants under the skin. Their mode of action is similar to that of steroidal oral contraceptive pills and their effective periods are much longer. Saheli is a non-steroidal oral contraceptive pill.

